Demonstration of Proposed Features

Date	13 July 2026
Team ID	SL-2026-02
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document records all the features proposed during the planning phase of the **Smart Lender – Loan Eligibility Prediction Using Machine Learning** project and verifies their successful implementation and demonstration. Each feature was developed, integrated, and tested to ensure that the final solution meets the functional requirements of an intelligent loan eligibility prediction system. The project demonstrates the effective application of Machine Learning and Flask-based web development to automate the loan approval prediction process.

S.No	Feature Name	Description	Status	Demonstrated	Remarks
1	Applicant Information Form	Allows users to enter loan applicant details such as income, education, credit history, loan amount, and property area through a responsive web interface.	**Implemented**	**Yes**	Successfully accepts and validates user input.
2	Data Preprocessing	Cleans, encodes, and transforms applicant data before prediction using preprocessing techniques.	**Implemented**	**Yes**	Ensures accurate model input and improves prediction reliability.
3	Machine Learning Prediction Model	Predicts loan eligibility using the trained Machine Learning model.	**Implemented**	**Yes**	Successfully integrated with the Flask application using Joblib.
4	Loan Eligibility Result	Displays whether the applicant is **Eligible** or **Not Eligible** based on the prediction result.	**Implemented**	**Yes**	Prediction results are generated instantly after submission.

5	Flask Web Application	Connects the frontend with the Machine Learning model and manages user requests.	**Implemented**	**Yes**	Application runs successfully on the local Flask server.
6	Input Validation & Error Handling	Validates user inputs and prevents invalid or incomplete data from being processed.	**Implemented**	**Yes**	Improves application reliability and user experience.

Feature Implementation Summary:

Metric	**Value**
Total Features Proposed	8
Total Features Implemented	8
Total Features Demonstrated	8